Practical Exercise Classification 2026 - Simulated Rocket Physics

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0.1 Motion

$$\dot{x} = v_x, \qquad \dot{y} = v_y$$

$$\dot{\mathbf{v}} = \frac{\mathbf{F}_{\text{thrust}} + \mathbf{F}_{\text{drag}} + \mathbf{F}_{\text{lift}}}{m} + \begin{pmatrix} 0 \\ -g \end{pmatrix}$$

where $g = 9.80665$

0.2 Thrust Direction

$$\mathbf{F}_{\text{thrust}} = T \begin{pmatrix} -\sin(\theta + \delta) \\ \cos(\theta + \delta) \end{pmatrix}$$

where θ is the rocket angle clockwise from vertical and δ is the gimbal angle.

0.3 Relative Air Velocity & Aero Forces

$$\mathbf{v}_r = \mathbf{v} - \mathbf{v}_{\text{wind}}, \qquad q = \frac{1}{2}\rho|\mathbf{v}_r|^2$$

$$\mathbf{F}_{\text{drag}} = -qAC_D\hat{\mathbf{v}}_r$$

$$\mathbf{F}_{\text{lift}} = qAC_L \begin{pmatrix} -\hat{v}_{r,y} \\ \hat{v}_{r,x} \end{pmatrix}$$

where density ρ is $1.225e^{-\frac{y}{8500}}$

0.4 Rotation

$$\dot{\theta} = \omega, \qquad \dot{\omega} = \frac{\tau_{\text{thrust}} + \tau_{\text{aero}}}{I}$$

$$\tau = \mathbf{r} \times \mathbf{F}$$

0.5 Fuel Consumption

$$\dot{m}_{\text{fuel}} = -\frac{T}{I_{sp} g_0}$$